

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

FinFETs, Nanosheets and Silicon-on-Insulator

KKKC6353 Nanoelectronic Devices — Lecture 4, Document 3 of 4

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The argument of this document

> **Recap from Lecture 1**
>
> Natural length: $\lambda = \sqrt{(\varepsilon_{Si}/\varepsilon_{ox})\, t_{Si}\, t_{ox}}$ for one gate; divide λ^2 by about 2 for a double gate and 4 for gate all around. Keep $L \gtrsim 5\lambda$.
>
> With 1 nm EOT this gives minimum gate lengths of about 39 nm (planar bulk), 15 nm (FinFET, t_{Si} = 6 nm) and 10 nm (nanosheet, t_{Si} = 5 nm).
>
> The four levers: thinner t_{ox}, shallower junctions and halos, thinner body (SOI), more gates (FinFET, GAA).

Lecture 1 explained *why* silicon transistors stopped being flat. The drain's field reaches into the channel over a natural length λ, and there are only two geometric ways to shorten λ: make the body thinner, and add gates. Silicon-on-insulator thins the body; the FinFET and the gate-all-around nanosheet add gates. Lecture 1, Document 1 told the industrial history, from the 22 nm FinFET of 2011 to the 2 nm nanosheets of 2025.

This document is about *how* those devices are designed and built, and about the physics that is peculiar to them. It makes four points:

1. A multi-gate transistor is specified by its **geometry**, not its doping. Its width is $2H_{fin} + W_{fin}$ per fin, comes in whole fins, and its electrostatics are set by the ratio of gate length to fin width. The nanosheet removes the whole-fin restriction.
2. In thin, undoped bodies the electrons are not at the surface: **volume inversion** and **quantum confinement** move them into the body and shift the threshold. These are design parameters, not curiosities.
3. The price of three-dimensional channels is **parasitic resistance, parasitic capacitance and self-heating**. At advanced nodes these, not the channel, decide performance.
4. **Silicon-on-insulator** is both a wafer technology and a device physics. Its partially depleted form has a floating body that causes the kink effect and history effects. Its fully depleted form has front–back gate coupling, which today's FD-SOI exploits as a tuning knob. SOI is not obsolete: in 2026 it is in most smartphone radio switches, low-power microcontrollers, and silicon photonics.

1 Multi-gate geometry

> **Key ideas**
>
> A FinFET is a double-gate (or tri-gate) transistor turned on its side: a thin vertical fin of silicon with the gate draped over it.
>
> Its effective width is $N_{fin}(2H_{fin} + W_{fin})$, so it comes in whole fins. Tall, closely spaced fins pack more width into each µm of layout.
>
> Electrostatics are set by L_g/W_{fin}: keep $L_{eff}/W_{fin} \gtrsim 1.5$.

1.1 From the double gate to the fin

The first double-gate transistor built in a standard process was Hitachi's **DELTA** (depleted lean-channel transistor), reported by Hisamoto and co-workers in 1989. A thin vertical wall

of silicon stood up from the substrate and the gate wrapped over it, so the two sides of the wall acted as two gates (Figure 3.1). The name **FinFET** came ten years later, from Hisamoto, Hu, King, Bokor and co-workers at Berkeley (1999–2000). They made the fin on SOI and self-aligned the gate to it. In 2002 the Berkeley group showed working FinFETs with a 10 nm gate length on a 12 nm fin (Figure 3.2).

There are three ways to build a double gate:

- **Planar double gate**: one gate above and one below a horizontal film. The best electrostatics, but the bottom gate is very hard to align to the top one.
- **Vertical channel**: current flows vertically, gates on the sides.
- **FinFET**: current flows horizontally along a vertical fin, gates on the two sidewalls (and the top, making a tri-gate). The gates are self-aligned because they are one piece of material. This is the one that won, because it is closest to a planar CMOS process.

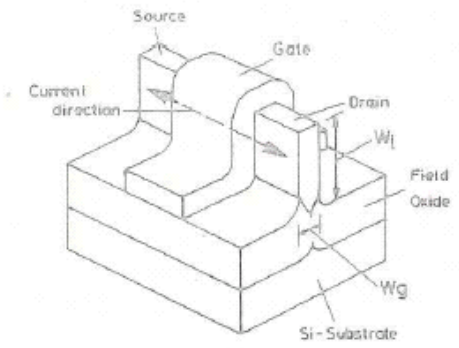


Figure 3.1. *The first FinFET-type device, Hitachi's DELTA: the channel is a vertical wall of silicon of width W_g, and the gate wraps over it. Source: D. Hisamoto, T. Kaga, Y. Kawamoto and E. Takeda, IEEE Electron Device Letters 11, 36 (1990); reproduced from Q. Zhang, EE 666 slides (2005).*

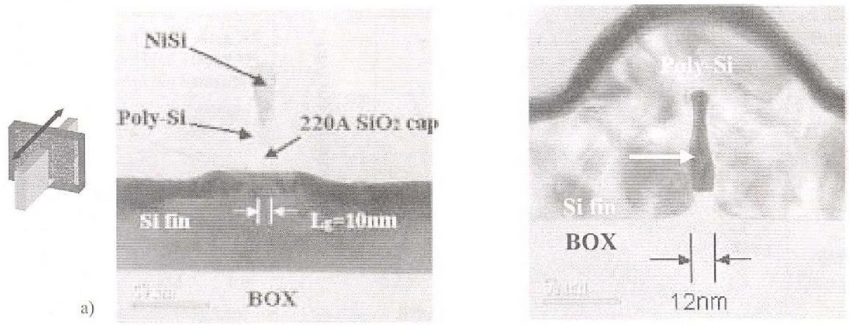


Figure 3.2. *The Berkeley FinFET of 2002, in cross-sections across the gate (left, L_g = 10 nm) and across the fin (right, fin width 12 nm), both on buried oxide. Source: B. Yu, C. Hu et al., IEDM 2002; reproduced from Q. Zhang, EE 666 slides (2005).*

1.2 Effective width

Current flows along the fin sidewalls and its top, so the effective width of one fin is

$$\boxed{W_{eff} = 2H_{fin} + W_{fin}}$$

and a device with N_{fin} fins has $N_{fin}(2H_{fin} + W_{fin})$. (In a double-gate FinFET with a thick hard mask on top, only the sidewalls conduct and $W_{eff} = 2H_{fin}$.) The effective channel

length is $L_{eff} = L_g + 2L_{ext}$, where L_{ext} is the undoped extension between the gate edge and the doped source or drain (Pei *et al.*, 2002).

Two consequences follow, both shown in Figure 3.4:

- **Width per footprint.** A fin of height H on a pitch P gives $(2H + W)/P$ µm of channel per µm of layout. With tall fins and tight pitches this exceeds 1. A FinFET packs *more* width into the same area than a planar transistor, which is how FinFET nodes increased drive current per µm while shrinking.
- **Fin quantisation.** Width comes only in whole fins. A designer cannot make a transistor 1.5 fins wide, so ratios in circuits such as SRAM cells, which need particular strength ratios, are restricted to ratios of small integers.

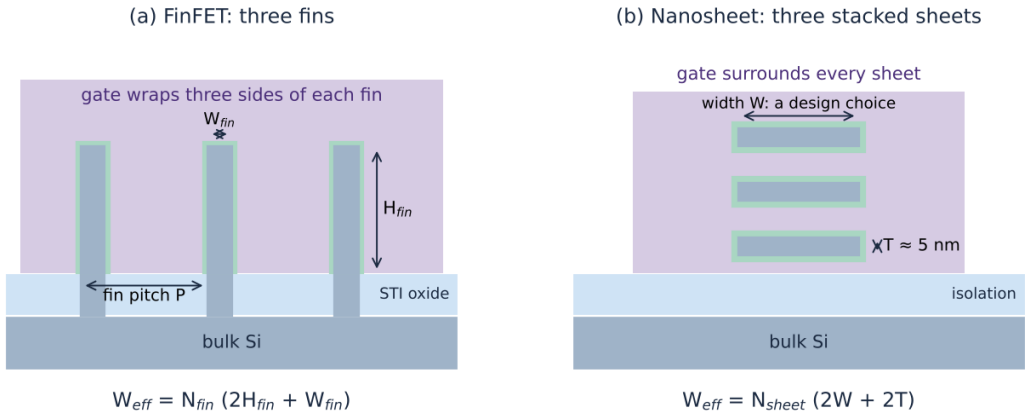


Figure 3.3. *Cross-sections across the gate. (a) A three-fin FinFET: the gate covers both sidewalls and the top of each fin. (b) A three-sheet gate-all-around nanosheet: the gate surrounds each sheet, and the sheet width is chosen freely. Figure made for this course; not to scale.*

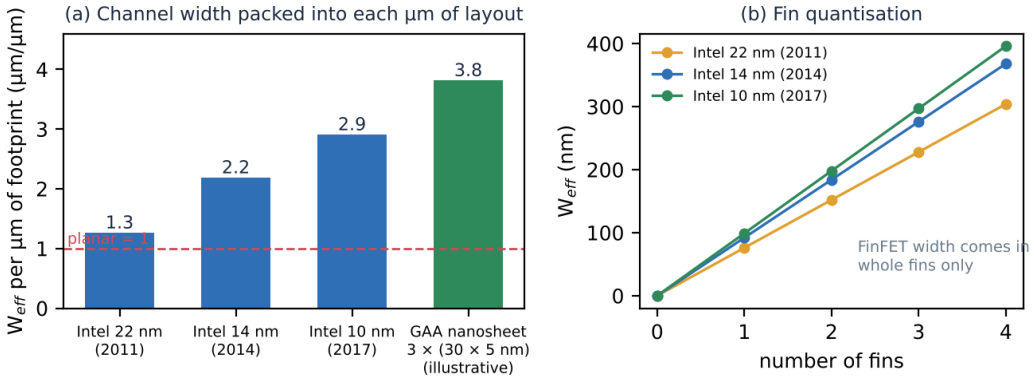


Figure 3.4. *(a) Effective channel width per µm of layout for three Intel FinFET generations, from published fin height, width and pitch, and for an illustrative three-sheet nanosheet stack. (b) FinFET width comes in multiples of $2H_{fin} + W_{fin}$. Figure made for this course; fin dimensions from Intel's IEDM papers (22 nm: H = 34, W = 8, P = 60 nm; 14 nm: 42, 8, 42 nm; 10 nm: 46, 7, 34 nm).*

> **Worked example: why fins got taller**
>
> At 22 nm (fin height 34 nm, width 8 nm, pitch 60 nm) one fin gives $W_{eff} = 76$ nm, i.e. $76/60 = 1.3$ µm per µm of layout. At 10 nm (46 nm, 7 nm, 34 nm) it gives $99/34 = 2.9$ µm per µm. More than doubling the width per footprint let Intel use *fewer fins* per

transistor (often two instead of three or four), which cut capacitance and area while keeping drive current. The same argument, carried to its end, gives the nanosheet: three sheets 30 nm wide and 5 nm thick give $3 \times 70 = 210$ nm of width from one stack.

1.3 Design rules for the fin

Pei and co-workers (2002) and Kim and co-workers (2004) worked out, by simulation, how the geometry sets the short-channel behaviour. These rules are in the old FinFET notes; they are summarised here:

- **Fin width.** Thinner fins suppress threshold roll-off and DIBL, as the natural length predicts. The rule of thumb is $L_{eff}/W_{fin} \gtrsim 1.5$. But thin fins are hard to make uniformly, and a 1 nm variation in a 6 nm fin shifts the threshold noticeably.
- **Fin height.** Taller fins give more width, and once the fin is taller than a critical value (which grows with fin width), roll-off and swing stop depending on height. A height of 20–90 nm is used in practice.
- **Corners.** The gate field concentrates at the top corners of a rectangular fin, which can turn on before the sidewalls and give a kinked subthreshold curve. Hydrogen annealing rounds the corners.
- **Threshold voltage.** Because the fin is undoped (Section 2), the threshold is set by the **gate work function**, with multiple work-function metals for different thresholds. Doping the fin to set the threshold, as in planar devices, brings back the variability FinFETs were meant to remove.

Check your understanding

1. Compute W_{eff} per μm of layout for a fin 50 nm tall, 6 nm wide, on a 30 nm pitch.
2. Why is fin quantisation a problem for SRAM design, and how does the nanosheet help?

2 Inside the fin: undoped bodies, volume inversion and confinement

Key ideas

Multi-gate channels are undoped. That removes impurity scattering and random dopant fluctuation, and it means the gates, not the doping, control the body.

In a thin fin the two gates' inversion layers merge into one in the middle of the body: volume inversion.

Below about 5 nm, quantum confinement raises the lowest subband and so raises the threshold voltage: a further source of sensitivity to fin width.

2.1 The undoped body

In a planar MOSFET the channel doping does two jobs: it sets the threshold and it limits how far the drain field penetrates. In a fully depleted thin body the geometry does the second job, so the doping is no longer needed for it. Leaving the fin undoped has three benefits, all listed in the course's 2026 FinFET slides:

- **No impurity scattering**, so higher mobility at a given charge density.

- **No random dopant fluctuation (RDF).** Lecture 1, Document 3 showed RDF to be the main source of threshold variation in planar devices. An undoped fin removes it, and the remaining variation comes from fin-width variation, work-function granularity in the metal gate, and line-edge roughness.
- **Lower vertical field** at a given charge, which also helps mobility.

2.2 Volume inversion

With two gates on a thick body, each gate forms its own inversion layer at its own surface. As the body thins, the two layers overlap, and below roughly 10 nm they merge into one, centred in the body (Figure 3.5). This is **volume inversion**, predicted for double-gate SOI by Balestra and co-workers (1987). Its effects are:

- **Higher mobility.** Electrons away from the interface scatter less on surface roughness and on charges in the gate dielectric.
- **A charge centroid away from the surface**, which slightly *lowers* the effective gate capacitance (the electrons are further from the gate).

The 2026 FinFET slides say volume inversion "further enhances current". That is right through the mobility, but it is a modest effect at today's channel lengths, where transport is quasi-ballistic and the gate capacitance matters more.

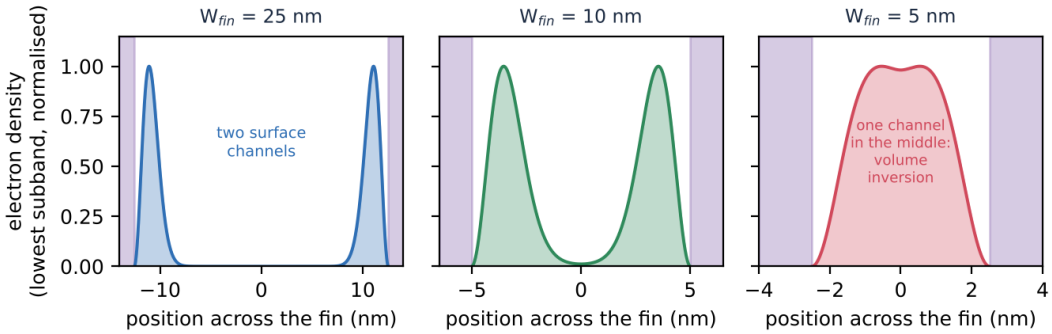


Figure 3.5. *The lowest-subband electron density across a double-gate fin, from a numerical solution of Schrödinger's equation in the potential of two gates. In a 25 nm fin there are two separate surface channels. At 10 nm they overlap. At 5 nm they have merged into one channel in the middle of the fin: volume inversion. Figure made for this course; effective mass 0.92 m_0, schematic field.*

2.3 Quantum confinement and the threshold

A fin of width W is a quantum well (Lecture 2). The lowest subband lies above the conduction-band edge by roughly

$$E_1 \approx \frac{\hbar^2 \pi^2}{2 m_\perp^* W^2}$$

where $m_\perp^*$ is the effective mass across the fin. For $W = 5$ nm this is about 16 meV with $m_\perp^* = 0.92\, m_0$, and about 80 meV with $m_\perp^* = 0.19\, m_0$; which mass applies depends on the fin's crystal orientation. The threshold rises by the same amount, and because $E_1 \propto 1/W^2$, a 1 nm change in a 5 nm fin shifts it by tens of millivolts. This is one reason fins stopped getting thinner at about 5–7 nm, and why nanosheets are about 5 nm thick.

2.4 Orientation and the two carriers

On a standard (100) wafer with fins along ⟨110⟩, the fin *sidewalls*, which carry most of the current, are (110) planes. Holes are faster on (110) than on (100); electrons are slower. The Berkeley p-FinFET of 2002 reached a peak transconductance of 633 µS/µm for exactly this reason. FinFET processes then add strain, with SiGe source and drain for pMOS and other stressors for nMOS, to recover electron mobility and boost holes further.

Check your understanding

1. Give three reasons for leaving the fin undoped.
2. What is volume inversion, and does it increase or decrease the gate capacitance?
3. Estimate the threshold shift when a 6 nm fin is thinned to 5 nm, taking $m_{\perp}^{*} = 0.19\, m_0$.

3 Making fins and nanosheets

> **Key ideas**
>
> Fins are narrower and more closely spaced than lithography can print. They are defined by spacer patterning: the fin is the thickness of a deposited film, not the width of a printed line.
>
> A nanosheet starts as a stack of alternating Si and SiGe layers. The SiGe is etched away after the fin is formed, and the gate is deposited into the gaps by atomic-layer deposition.
>
> The FinFET was "easy in concept, tough to build"; the nanosheet is harder still.

3.1 Spacer patterning

A fin 7 nm wide on a 34 nm pitch cannot be printed directly, even by EUV lithography. Instead a sacrificial line ("mandrel") is printed and etched, a thin film is deposited over it conformally, and the film is etched back so it remains only on the mandrel's sidewalls as **spacers**. The mandrel is removed and the spacers become the etch mask for the fins. Each mandrel gives two fins, so the fin pitch is half the printed pitch, and the fin *width* is set by the deposited film thickness, which is controlled to the ångström. Repeating the trick (self-aligned quadruple patterning, SAQP) quarters the pitch.

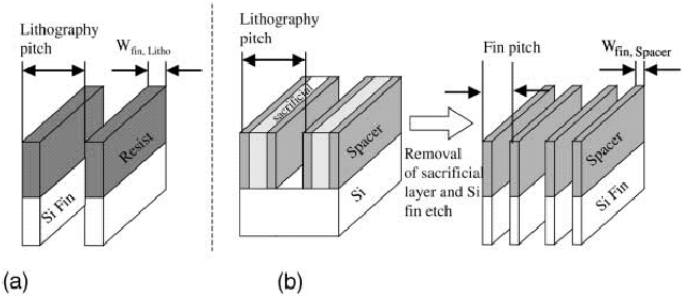


Fig. 3. Comparison of fin density achieved with (a) conventional lithography and (b) spacer lithography technology. The spacer lithography technology produces twice the fin density of a conventional lithography technology.

Figure 3.6. *Fin density from conventional lithography (a) and from spacer lithography (b): removing the sacrificial layer leaves two spacers per printed line, doubling the fin density, and the fin width is set by the spacer thickness. Source: after Y.-K. Choi et al., Solid-State Electronics 46, 1595 (2002), and W. Rösner et al., Solid-State Electronics 48, 1819 (2004), as cited in Q. Zhang, EE 666 slides (2005), from which it is reproduced.*

3.2 The FinFET process

A bulk FinFET process then runs roughly as follows:

1. Pattern the fins by spacer patterning and etch them into the silicon.
2. Fill between the fins with oxide (shallow-trench isolation, STI) and recess it. **The depth of the recess sets H_{fin}.**
3. Form a sacrificial polysilicon "dummy" gate across the fins, then spacers.
4. Grow raised source and drain regions epitaxially on the exposed fins (SiGe:B for pMOS, Si:P for nMOS), which lowers resistance and adds strain.
5. Fill with dielectric, remove the dummy gate, and deposit the high-κ dielectric and work-function metals into the trench (**replacement metal gate**, the "gate-last" process of Lecture 1).

The Berkeley process of 2002 in the old notes used a SiN hard mask on SOI, a sacrificial oxidation to remove etch damage, a thin gate oxide and a polysilicon gate. Its logic is the same; every step has since been refined.

3.3 The nanosheet process

A gate-all-around nanosheet adds four steps to that flow (Figure 3.7):

- **A superlattice** of alternating Si and SiGe layers, a few nanometres each, is grown epitaxially before the fins are etched.
- After the dummy gate is removed, the SiGe is etched away selectively, **releasing** the silicon sheets so they hang between the source and drain.
- **Inner spacers** are formed in the recesses left at the ends of the SiGe layers, to keep the gate from touching the source and drain and to reduce capacitance.
- **Atomic-layer deposition** coats every sheet, top, bottom and sides, with high-κ dielectric and then metal, filling the gaps between sheets.

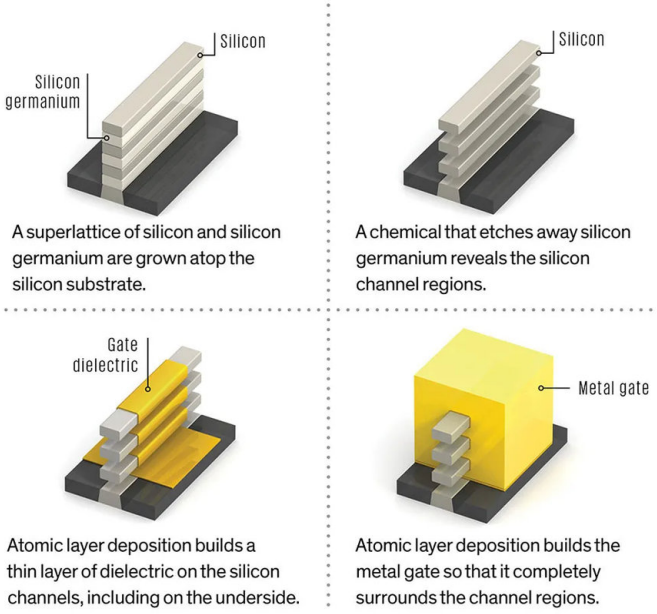


Figure 3.7. *From planar to FinFET to stacked nanosheet (left), and the nanosheet process (right): grow a Si/SiGe superlattice; etch away the SiGe to free the silicon sheets; coat them with dielectric by atomic-layer deposition; fill with metal gate. Source: P. Ye, T. Ernst and M. V. Khare, "The nanosheet transistor is the next (and maybe last) step in Moore's law", IEEE Spectrum (2019); reproduced from the 2026 KKKC6353 FinFET slides.*

Check your understanding

1. In spacer patterning, what sets the fin width and what sets the fin pitch?
2. Which step in a FinFET process sets the fin height?
3. Why does a nanosheet need inner spacers?

4 The price of three dimensions: parasitics and self-heating

> **Key ideas**
>
> Thin fins and sheets have high access resistance, and the 3D gate has more capacitance to the source and drain contacts. At advanced nodes these parasitics take a large share of the delay.
>
> Fins and sheets are surrounded by oxide, which conducts heat about 100 times worse than silicon, so the channel runs hot: self-heating.
>
> Self-heating lowers mobility and speeds up reliability failures (BTI, hot-carrier damage).

4.1 Parasitic resistance

Current must get from a metal contact into a 7 nm fin. The thin fin has high sheet resistance, and the contact area is small. The response is raised, heavily doped epitaxial source and drain regions that merge over neighbouring fins, and contact metals (silicides, then wrap-around contacts) with very low specific contact resistivity, of order $10^{-9}\ \Omega\ \text{cm}^2$. Even so, the source and drain series resistance is a large fraction of the total on-resistance of an advanced transistor. This is the same problem as the CNTFET contacts of Document 2, in a milder form.

4.2 Parasitic capacitance

The gate now faces the source and drain contacts across the fin's full height, and in a nanosheet also across the gaps between sheets. Fringing capacitance between gate and contact can exceed the intrinsic channel capacitance. Low-κ spacers, inner spacers and careful contact placement reduce it. A transistor's switching delay is roughly CV_{DD}/I_{on}, so the parasitic capacitance counts as much as the drive current.

4.3 Leakage through thin dielectrics

With equivalent oxide thicknesses below 1 nm, direct tunnelling through the gate dielectric (Lecture 3, Document 1) would be enormous with SiO_2. High-κ dielectrics (HfO_2, $\kappa \approx 20–25$) give the same capacitance with a physically thicker layer of 1.5–2 nm, which suppresses tunnelling. This was introduced at 45 nm (Lecture 1) and is essential for every FinFET and nanosheet.

4.4 Self-heating

A fin is surrounded by gate dielectric and STI oxide; an SOI fin also sits on buried oxide. Figure 3.8 shows why that matters: heat generated in the channel must leave through materials that conduct it 100 to 1000 times worse than bulk silicon, and the fin itself conducts worse than bulk because phonons scatter off its surfaces. The channel can run tens of kelvin above the substrate under continuous high current. The results are:

- **Lower mobility** and drive current, because phonon scattering rises with temperature.
- **Faster reliability degradation**: bias-temperature instability (NBTI in pMOS) and hot-carrier injection both accelerate exponentially with temperature (Lecture 1, Document 3).
- **Harder circuit modelling**, because the temperature depends on the switching history.

In logic, transistors switch only part of the time, so the average temperature rise is smaller than in DC measurements. Self-heating is most severe in analog and I/O transistors that carry steady currents, and in stacked nanosheets and SOI.

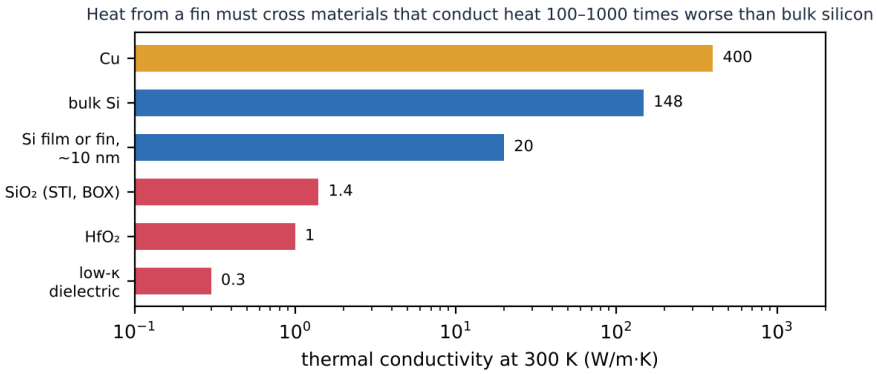


Figure 3.8. *Thermal conductivities of the materials around a fin. Figure made for this course; typical room-temperature values from the literature; the thin-film value depends strongly on thickness.*

> **A correction to the old notes.** The course's 2026 FinFET slides attribute the high drive current of short FinFETs to "velocity overshoot: … quasi-ballistic transport (carriers traverse the channel without scattering)". These are two different things. *Velocity overshoot* means carriers briefly exceed the saturation velocity because they have not yet had time to scatter and lose energy. *Quasi-ballistic transport* means some carriers cross with few or no scattering events at all. Both occur in short channels and both raise the current, but they are distinct (Lecture 1, Document 2, Section 4.1).

Check your understanding

1. Why does the parasitic capacitance matter as much as the drive current for switching speed?
2. Why is self-heating worse in SOI and in nanosheets than in bulk FinFETs?

5 Silicon-on-insulator wafers

> **Key ideas**
>
> An SOI wafer has a thin layer of crystalline silicon on a buried oxide (BOX) on a silicon handle wafer.
>
> Silicon-on-sapphire came first (1960s); SIMOX implants the oxide; bonded wafers join two wafers; Smart Cut (1995) splits a bonded wafer along an implanted hydrogen layer and is how almost all SOI is made today.
>
> Today's FD-SOI wafers have a 5–7 nm silicon film on a 20–25 nm BOX, uniform to a few ångströms across a 300 mm wafer.

5.1 Silicon-on-sapphire

The first SOI material was silicon grown epitaxially on sapphire (Al_2O_3), introduced in the 1960s. Sapphire is an excellent insulator and radiation-hard, which made SOS the choice for space and military circuits. But the lattice mismatch leaves the silicon full of defects near the interface, and electron mobility in SOS MOSFETs is only about 230–250 $cm^2/V\,s$. SOS survives today in some RF switches and radiation-hardened parts.

5.2 SIMOX

SIMOX (separation by implanted oxygen; Izumi *et al.*, 1978) implants a very high dose of oxygen ions, about $0.3–1.8 \times 10^{18}$ cm^{-2} at 120–200 keV, into a silicon wafer, then anneals it above 1300 °C for several hours so the oxygen forms a buried SiO_2 layer below a surface layer of crystalline silicon. Typical BOX thicknesses were 100–400 nm and silicon films 50–240 nm. Early SIMOX had about 10^9 dislocations per cm^2 in the silicon film. Multiple implants and better anneals brought that down to about 10^3 cm^{-2} by the late 1990s (Figure 3.9).

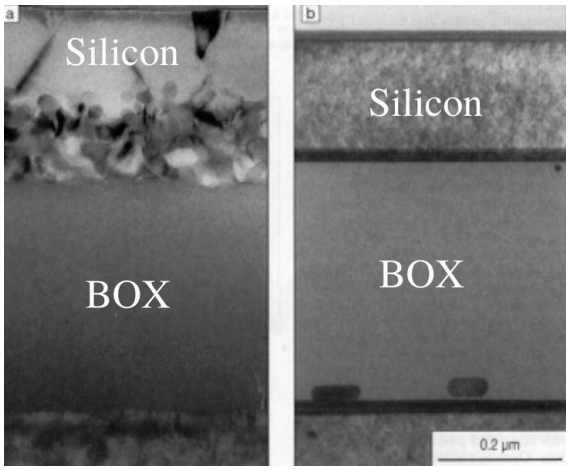


Figure 3.9. *SIMOX material in cross-section: about 1985 (left), with a heavily defected silicon film, and 1998 (right), with a clean film on a uniform buried oxide. Dislocation densities fell from about 10^9 to about 10^3 per cm^2. Source: J.-P. Colinge, "SOI technology: overview and device physics", NSREC 2001 short course (in the old notes).*

5.3 Bonding and Smart Cut

In **bonded and etched-back SOI (BESOI)**, an oxidised wafer is pressed against a second wafer. The two bond at room temperature through surface hydroxyl groups and van der Waals forces, and an anneal at about 1100 °C converts the bond to Si–O–Si. One wafer is then ground and etched back to the required thickness. Thickness control is the weakness: it is hard to etch a 725 µm wafer down to 50 nm uniformly.

Smart Cut (Bruel, 1995), commercialised as Unibond® by Soitec, solves that problem (Figure 3.10):

1. Oxidise wafer A and implant hydrogen ions through the oxide, to a dose of about 10^{16}–10^{17} cm^{-2}. The hydrogen collects at a depth set by the implant energy.
2. Clean both wafers and bond A face-down onto a handle wafer B.
3. Heat to 400–600 °C. The hydrogen forms microcavities that join up, and A splits off along the implanted layer, leaving a thin film of silicon on B.
4. Anneal at about 1100 °C to strengthen the bond, and polish the surface.

The film thickness is set by the implant energy and the oxide thickness, both controlled very precisely. The leftover wafer A is polished and reused as the next donor, which keeps the cost down. Almost all SOI wafers made today are made this way. A related process, ELTRAN (Canon), splits at a porous-silicon layer.

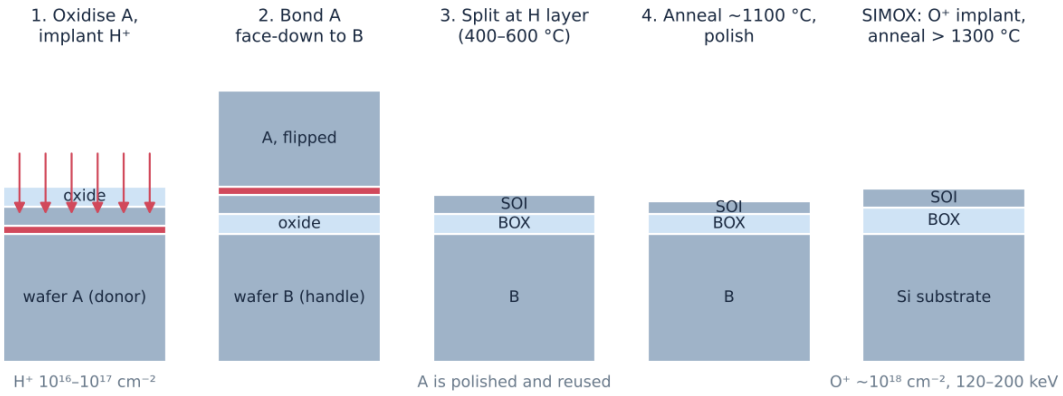


Figure 3.10. *Smart Cut in four steps, and SIMOX for comparison. Figure made for this course; not to scale.*

Check your understanding

1. Why does SIMOX need such a high implant dose and such a hot anneal?
2. What sets the thickness of the silicon film in Smart Cut, and why does that give better control than etch-back?

6 Partially and fully depleted SOI MOSFETs

> **Key ideas**
>
> If the silicon film is thicker than the maximum depletion depth, a neutral, floating body remains under the channel: partially depleted (PD) SOI. If it is thinner, the whole film is depleted: fully depleted (FD) SOI.
>
> The floating body of PD-SOI collects holes from impact ionisation; this raises the body potential, lowers V_T, and causes the kink effect and history effects.
>
> In FD-SOI the front and back interfaces are coupled through the film. The back gate (the substrate) shifts the front threshold, which today's FD-SOI uses deliberately.

6.1 The condition

For body doping N_A, the maximum depletion depth under the gate (at threshold, surface potential $2\phi_F$) is

$$x_{d,max} = \sqrt{\frac{4\varepsilon_{Si}\phi_F}{qN_A}}, \qquad \phi_F = U_T \ln\frac{N_A}{n_i}$$

The device is **fully depleted** if $t_{Si} < x_{d,max}$: the depletion region reaches the BOX. It is **partially depleted** if $t_{Si} > x_{d,max}$; if the back interface is also depleted, the condition for a neutral region to survive is $t_{Si} > 2x_{d,max}$ (Figure 3.11).

> **A correction to the old notes.** The title of Bhat's slide on partially depleted SOI reads "$t_{Si} < 2x_{d(max)}$". The inequality is the wrong way round: a neutral body remains only if $t_{Si} > 2x_{d(max)}$, when the front and back depletion regions cannot meet. The slide's own drawing, with a neutral layer between two depletion regions, shows the correct case.
>
> **Worked example: PD or FD?**

For $N_A = 10^{17}$ cm^{-3}, $\phi_F = 0.0259\ln(10^{17}/10^{10}) = 0.42$ V and $x_{d,max} = \sqrt{4 \times 1.04 \times 10^{-12} \times 0.42/(1.6 \times 10^{-19} \times 10^{17})} = 104$ nm. A 150 nm film is partially depleted (the IBM processors of 2000–2010 used films of this order). A 50 nm film is fully depleted. A 7 nm film, as in 28 nm FD-SOI, is fully depleted at any reasonable doping, and is in fact left undoped.

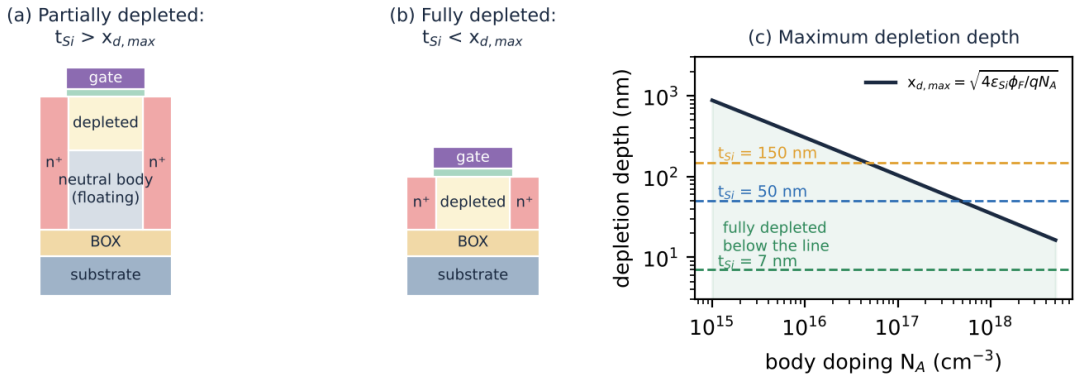


Figure 3.11. *(a) Partially depleted: a neutral, floating body remains under the depletion region. (b) Fully depleted: the whole film is depleted. (c) The maximum depletion depth against doping, with three film thicknesses: below the line, the film is fully depleted. Figure made for this course.*

6.2 The floating body and the kink effect

In a PD-SOI nMOSFET the neutral body is connected to nothing. Colinge's tutorial in the old notes sets out the kink effect as a six-step chain:

1. At high drain voltage, **impact ionisation** near the drain creates electron–hole pairs (Lecture 1, Document 3).
2. The electrons go to the drain; the **holes flow into the floating body**, which, unlike a bulk body, has no substrate contact to drain them away.
3. The body charges up until the **body–source diode** is slightly forward-biased and passes the excess holes to the source.
4. The body potential has **risen** by a few tenths of a volt.
5. Through the body effect, the **threshold voltage falls**.
6. The **drain current jumps up**: a "kink" in the output characteristics (Figure 3.12).

The extra current might look useful, but the kink makes the output conductance poor and bias-dependent, which is bad for analog circuits, and it makes the current depend on how long the device has been on. Related **floating-body effects** include:

- **History effects**: the body charge, and so V_T, depends on the recent switching history, so the delay of a logic gate varies from cycle to cycle.
- **The parasitic bipolar transistor**: source (emitter), body (base) and drain (collector) form a lateral npn. With the base floating, it can amplify the impact-ionisation current.
- **Single-transistor latch**: at high drain bias the bipolar feedback can make the subthreshold swing infinitely steep and the device impossible to turn off.

The cure is a **body tie**, a p^+ contact to the body (for example an H-shaped gate with body contacts at each end). It costs area, so IBM's PD-SOI processors tied only the few critical transistors and designed the rest of the circuits to tolerate the floating body.

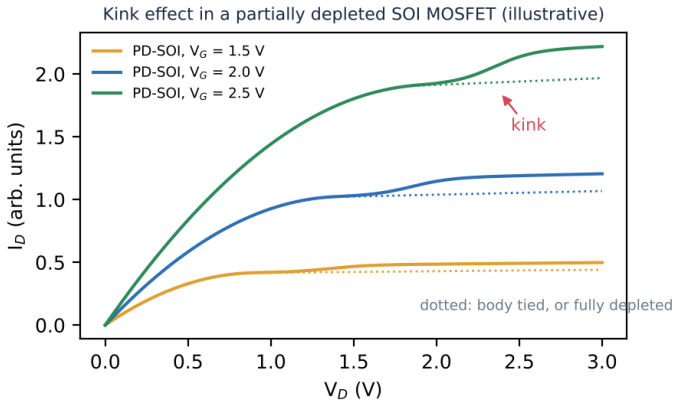


Figure 3.12. *The kink effect in a partially depleted SOI nMOSFET (solid), against the same device with a body tie or in fully depleted form (dotted). Figure made for this course; illustrative.*

> **A correction to the old notes.** The Microwind chapter (Sicard and Delmas-Bendhia, 2003) credits the kink effect with making the SOI current "20% higher", as a performance gain. The kink is a parasitic effect. It raises the current only in part of the output curve, degrades the output resistance, and makes timing history-dependent. Fully depleted devices were developed precisely to remove it.

6.3 Fully depleted SOI: front–back coupling

In FD-SOI the depletion region spans the film, so the front and back interfaces are electrostatically coupled. Each interface can be accumulated, depleted or inverted, giving nine combinations (the operation-mode map in Bhat's slides). In normal operation the back interface is depleted, and the front threshold depends linearly on the back-gate (substrate) voltage. Lim and Fossum (1983) derived

$$\boxed{\frac{\partial V_{Tf}}{\partial V_{Gb}} = -\frac{C_{Si}\,C_{ob}}{C_{of}\,(C_{Si}+C_{ob})}}$$

with C_{of} and C_{ob} the front and back oxide capacitances and $C_{Si} = \varepsilon_{Si}/t_{Si}$ the film capacitance. For a thin film, $C_{Si} \gg C_{ob}$, and the coefficient is about $C_{ob}/C_{of} = t_{of}/t_{ob}$, the ratio of the oxide thicknesses. Once the back interface inverts or accumulates, it screens the back gate and V_{Tf} stops moving (Figure 3.13).

Two consequences:

- **Near-ideal subthreshold swing.** The body factor is $n = 1 + C_{Si}C_{ob}/\left[C_{of}(C_{Si} + C_{ob})\right]$, about 1.05 in FD-SOI against about 1.5 in bulk, so the swing is about 62 mV/dec instead of 90 mV/dec. A steeper swing means a lower threshold for the same leakage, and so a lower supply voltage. (The Microwind chapter's "NFACT" of 1.0 for SOI against 1.5 for bulk is the same body factor.) This applies to **fully** depleted devices; Bhat's list of SOI advantages gives "reduced subthreshold swing" for SOI in general, but a PD device has the swing of a bulk device.

- **Back-gate tuning.** With a thin BOX, the substrate becomes a usable second gate. Modern FD-SOI (ultra-thin body and BOX, UTBB) uses a 20–25 nm BOX and shifts V_T by about 80 mV per volt of back bias: forward body bias for speed, reverse body bias for low leakage, set per circuit block or even dynamically.

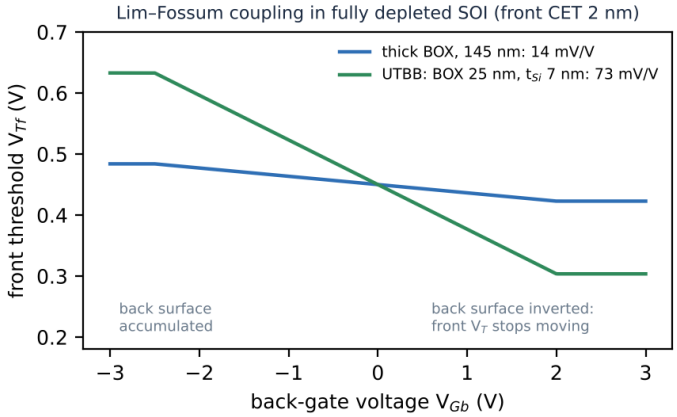


Figure 3.13. *Front threshold against back-gate voltage in an FD-SOI MOSFET, from the Lim–Fossum expression with a 7 nm film and a front CET of 2 nm. A thick BOX couples weakly; a 25 nm BOX turns the substrate into a usable second gate. Figure made for this course.*

Check your understanding

1. Is a 70 nm film doped at 3×10^{17} cm^{-3} partially or fully depleted?
2. List the six steps of the kink effect.
3. Why does a thin BOX make the back gate useful, and what does it cost in junction capacitance (Section 7.1)?

7 What SOI buys, and what it costs

> **Key ideas**
>
> SOI isolates every transistor in oxide. That removes most junction capacitance, latch-up and substrate noise, and shrinks the volume in which radiation can deposit charge.
>
> It costs a more expensive wafer, self-heating, floating-body effects (in PD-SOI) and a weakness to total-dose radiation in the buried oxide.

7.1 Junction capacitance

In bulk CMOS the source and drain sit in the body and have junction capacitance to it over their bottom and sidewalls. In SOI their bottom faces the BOX and their sides face isolation oxide, so most of that capacitance disappears.

> **Worked example: drain capacitance, bulk against SOI**
>
> Take the SPICE parameters for the NMOS model in the Carleton notes: $C_J = 5.9 \times 10^{-4}$ F/m^2 (0.59 fF/μm^2), grading coefficient $M_J = 0.767$, built-in potential $P_B = 0.99$ V. For a drain region 10 μm by 6 μm (60 μm^2) at 2.5 V reverse bias: $C_{db} = 0.59 \times 60/(1 + 2.5/0.99)^{0.767} = 35.4/2.63 = 13.5$ fF, plus a small sidewall term.
>
> On SOI with a 145 nm BOX, the drain bottom sees the BOX capacitance ($3.9\varepsilon_0/145$ nm $= 0.24$ fF/μm^2) in series with the depletion capacitance of a lightly doped

> substrate (about 0.1 fF/μm² at 10^{15} cm^{-3}), giving about 0.07 fF/μm², or 4.5 fF for the same area: three times smaller at this bias, and the sidewall term is gone. With a 25 nm UTBB BOX the BOX capacitance is 1.4 fF/μm², so the advantage shrinks; there the back gate is the point (Section 6.3).

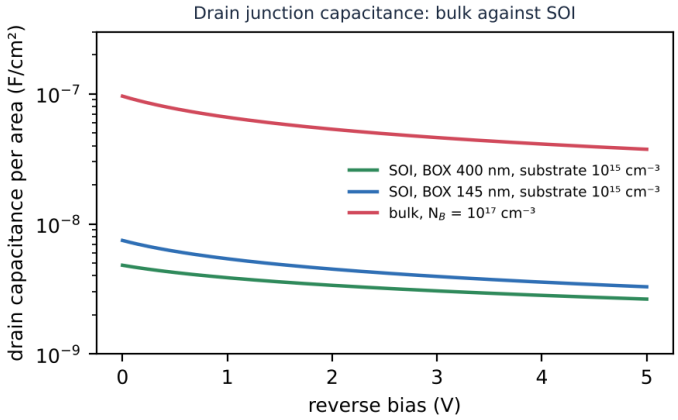


Figure 3.14. *Drain capacitance per unit area against reverse bias for a bulk junction on* 10^{17} *cm*$^{-3}$ *and for SOI drains on two BOX thicknesses over a lightly doped substrate. Figure made for this course, after the comparison in Colinge's tutorial.*

Lower capacitance means faster switching at the same current, or the same speed at a lower supply voltage. The Microwind simulations in the old notes put numbers on this for a 0.12 μm process (2003): a three-inverter ring oscillator ran at about 40 GHz in both technologies, but at 0.7 V and 66 μW in SOI against 1.2 V and 266 μW in bulk.

7.2 Isolation, density and latch-up

- **No latch-up.** Bulk CMOS contains a parasitic pnpn thyristor between the nMOS and pMOS wells, which a current spike can trigger, shorting the supply. In SOI each transistor sits in its own oxide-isolated island and there is no thyristor path.
- **Density.** Because n^+ and p^+ regions can abut without wells, the n-to-p spacing rule drops sharply; the Microwind rules go from 12λ in bulk to 2λ in SOI, making the ring oscillator 20% smaller.
- **Low substrate noise.** The BOX decouples noisy digital blocks from sensitive analog and RF circuits on the same chip, which is the main reason RF front ends use SOI.

7.3 Harsh environments

- **High temperature.** Junction leakage rises steeply with temperature and is proportional to junction area. With almost no junction area, SOI circuits work at 200–300 °C, where bulk circuits leak too much (the Microwind leakage plots show this).
- **Radiation: single events.** A heavy ion leaves a track of charge. In bulk silicon the charge from microns of track is collected; in a thin SOI film only the charge in the film is, so SOI is far less susceptible to single-event upsets.
- **Radiation: total dose.** The BOX itself traps positive charge under ionising radiation, which shifts the back interface and can turn on a back channel. Radiation-hardened SOI uses hardened BOX processes and body ties. SOI is not automatically radiation-hard.

7.4 Costs

The wafer is more expensive than bulk silicon; in 2003 the Microwind chapter put it at 5–10 times, and the gap has narrowed with Smart Cut volume, but it is still several times. The thermal penalty of the BOX was described in Section 4. PD-SOI's floating body complicates design and modelling, which is why it needed special compact models (BSIMSOI) and design rules.

Check your understanding

1. Explain why SOI eliminates latch-up.
2. Why does SOI help against single-event upsets but not automatically against total-dose radiation?

8 SOI and multi-gate devices in 2026

> **Key ideas**
>
> High-performance logic moved from PD-SOI (IBM, AMD, until the early 2010s) to bulk FinFETs and then to bulk-substrate nanosheets.
>
> FD-SOI with back-gate bias is in volume production at 28, 22 and 18 nm for low-power, IoT, automotive and RF chips.
>
> RF-SOI is in most smartphone radio front ends, and SOI is the standard platform for silicon photonics.

8.1 Where each technology went

- **PD-SOI for processors.** IBM used PD-SOI for its POWER and PowerPC processors from 2000, and AMD for its processors until 32 nm (2011). The Colinge tutorial's list of SOI users (IBM, Motorola, AMD, TI, HP, Peregrine, Honeywell) is a snapshot of about 2001. IBM later built FinFETs on SOI at 22 and 14 nm. The rest of the industry built FinFETs on bulk wafers, which are cheaper and conduct heat better, and that is also where the nanosheet went.
- **FD-SOI.** Planar, undoped, 5–7 nm films on 20–25 nm BOX, with back-gate bias. In volume at 28 nm (STMicroelectronics, Samsung) and 22 nm (GlobalFoundries 22FDX). In 2024 STMicroelectronics and Samsung Foundry announced an **18 nm FD-SOI** process with embedded phase-change memory for microcontrollers, sampling in the second half of 2024 and in production in the second half of 2025. FD-SOI is planar, so its masks and design costs are much lower than a FinFET's, and its back bias trades speed for leakage after manufacture.
- **RF-SOI.** Thick-film SOI on high-resistivity (often trap-rich) handle wafers is used for the antenna switches and low-noise amplifiers of almost every smartphone, replacing GaAs and SOS switches (Document 4).
- **Photonic SOI.** Silicon waveguides need a low-index cladding underneath; the BOX provides it, so silicon photonics for data-centre optical links is built on SOI wafers.

8.2 The convergence

The nanosheet can be seen as the end point of the SOI argument. Each sheet is a thin, undoped, fully depleted silicon film isolated from the substrate, with the gate on all four

sides instead of a buried oxide on one. The old notes treat SOI, the double gate and the FinFET as separate topics; physically they are one idea, as the multi-gate DIBL comparison in Colinge's tutorial already showed in 2001 (Figure 3.15).

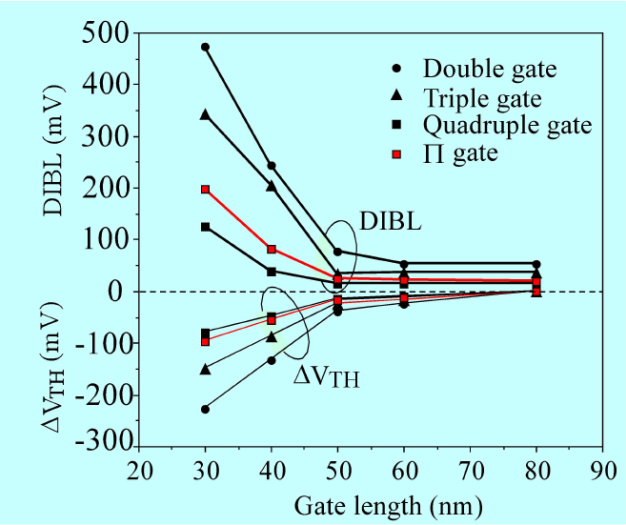


Figure 3.15. *DIBL and threshold shift against gate length for fully depleted SOI devices with double, triple, quadruple and Π-shaped gates: more gate coverage, less DIBL. Source: J.-P. Colinge, NSREC 2001 short course (in the old notes).*

Check your understanding

1. Why did high-performance logic leave SOI?
2. Give two products in 2026 that rely on SOI, and say what SOI does for each.

9 Summary

- A FinFET is a self-aligned double or tri-gate device on a vertical fin: $W_{eff} = N_{fin}(2H_{fin} + W_{fin})$, $L_{eff} = L_g + 2L_{ext}$; keep $L_{eff}/W_{fin} \gtrsim 1.5$. Tall fins on tight pitches gave 2–3 µm of width per µm of layout. Nanosheets remove fin quantisation.
- Fins and sheets are undoped: no RDF, higher mobility, V_T set by the gate work function. Below about 10 nm the two inversion layers merge (volume inversion), and below about 5 nm quantum confinement shifts V_T as $1/W^2$.
- Fins are made by spacer patterning, nanosheets from Si/SiGe superlattices by selective release, inner spacers and ALD gate stacks.
- Three dimensions bring parasitic resistance, parasitic capacitance and self-heating, which now take a large share of the performance budget.
- SOI wafers: SOS, SIMOX, bonding, and Smart Cut (most of today's wafers).
- PD-SOI ($t_{Si} > x_{d,max}$; $2x_{d,max}$ with back depletion) has a floating body: kink, history effects, parasitic bipolar, single-transistor latch; cured by body ties. FD-SOI ($t_{Si} < x_{d,max}$) has front–back coupling, $\partial V_{Tf}/\,\partial V_{Gb} \approx -t_{of}/t_{ob}$, a body factor near 1, and usable back bias.
- SOI cuts junction capacitance, prevents latch-up, isolates noise and resists single-event upsets; it costs wafer price and heat removal.
- In 2026: logic on bulk nanosheets; FD-SOI at 28–18 nm for low power; RF-SOI in phones; SOI for photonics.

Corrections to the old notes

Old note	What it said	Correction
Bhat (IISc), slide 20	Partially depleted SOI: "$t_{Si} < 2x_{d(max)}$"	$t_{Si} > 2x_{d(max)}$
Bhat, slide 17	SOI in general has a "reduced sub-threshold swing"	Only fully depleted SOI; PD-SOI has a bulk-like swing
Sicard & Delmas-Bendhia (Microwind, 2003)	Kink effect gives 20% more current, presented as a benefit	A parasitic effect; it degrades output resistance and timing
Sicard & Delmas-Bendhia	Draft with placeholders ("add an example", "tbd", "Figure 15-xxx")	Used here only as a source of numbers
Colinge (2001)	Industry snapshot: IBM PowerPC G4/G5, Motorola, HP…	See Section 8 for the 2026 picture
KKKC6353 FinFET slides (2026)	Velocity overshoot equated with quasi-ballistic transport	Two related but distinct effects (Section 4.4)
KKKC6353 FinFET slides	Effective width not given	$W_{eff} = 2H_{fin} + W_{fin}$ per fin
Zhang (2005)	"Smallest in 2002: 6.5 nm Si fin"	Historical; production fins reached about 7 nm and nanosheets about 5 nm

Words to know

Term	Meaning
FinFET / tri-gate	A transistor whose channel is a vertical fin with the gate on two or three sides
Fin quantisation	FinFET width comes only in whole fins
Volume inversion	Merging of the two inversion layers into one in the middle of a thin body
Spacer patterning (SADP/SAQP)	Defining features by deposited sidewall spacers, halving or quartering the printed pitch
Inner spacer	Dielectric between the gate and source/drain in the gaps between nanosheets
Self-heating	Heating of the channel by its own current, made worse by surrounding oxide
BOX	Buried oxide of an SOI wafer
Smart Cut	Splitting a bonded wafer along an implanted hydrogen layer to make SOI
PD-SOI / FD-SOI	Partially or fully depleted SOI film
Floating-body effects	Kink, history effects and bipolar action caused by charge stored in an unconnected body
Lim–Fossum coupling	Dependence of the front threshold on the back-gate voltage in FD-SOI
UTBB	Ultra-thin body and BOX: FD-SOI with a 20–25 nm BOX for back-bias control

In-class animations

Both at faizverse.github.io/nanoelectronic-devices/L4.

- **FinFET and nanosheet width.** Set the fin height, width and pitch, or the number, width and thickness of nanosheets, and read off the effective width, the width per µm of layout and a natural-length estimate. Try reproducing the three Intel generations in Figure 3.4.
- **PD and FD SOI.** Set the film thickness and doping and watch the depletion region fill the film; move the back-gate voltage and watch the front threshold follow the Lim–Fossum line; switch on the kink in a PD device.

Sources and further reading

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- *MOSFET Capacitances*, Carleton University 97.477 lecture (2003).

- Q. Zhang, *FinFET*, EE 666 lecture slides (2005).
- *Introduction to FinFETs*, KKKC6353 slides (January 2026).